
Tutorial No.: 8
Engineering Mechanics (ME-101)
Date: 18/03/2024 and Time: 7.55 to 8.55 AM

Question 1: The 10-kg steel sphere is suspended from the 15-kg frame which slides down the 20° incline, as shown in Figure 1. If the coefficient of kinetic friction between the frame and incline is 0.15, compute the tension in each of the supporting wires A and B. Take $g = 9.81 \text{ m/s}^2$.

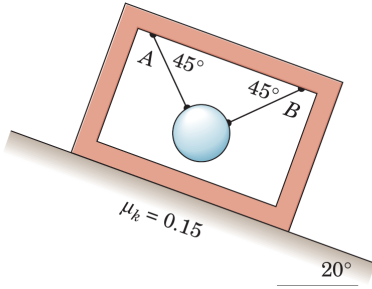


Figure 1

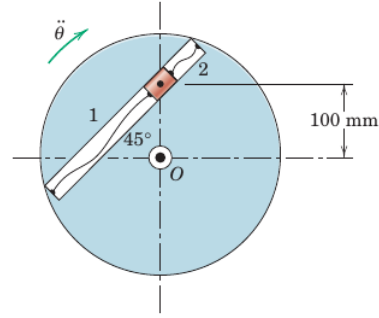


Figure 2

Question 2: The 2-kg slider fits loosely in the smooth slot of the disk, which rotates about a vertical axis through point O, as illustrated in Figure 2. The slider is free to move slightly along the slot before one of the wires becomes taut. If the disk starts from rest at time $t = 0$ and has a constant clockwise angular acceleration of 0.5 rad/s^2 , plot the tensions in wires 1 and 2 and the magnitude N of the force normal to the slot as functions of time t for the interval $0 \leq t \leq 5 \text{ s}$.

Question 3: The spring attached to the 10-kg mass is non-linear, having the force–deflection relationship shown in the Figure 3, and is unstretched when $x = 0$. If the mass is moved to the position $x = 100 \text{ mm}$ and released from rest, determine its velocity v when $x = 0$. Determine the corresponding velocity v' if the spring would have been linear according to $F = 4x$, where x is in meter and the force F is in kN.

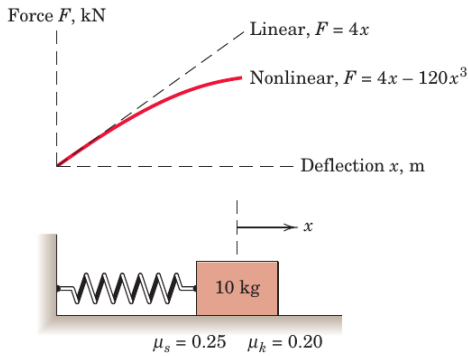


Figure 3

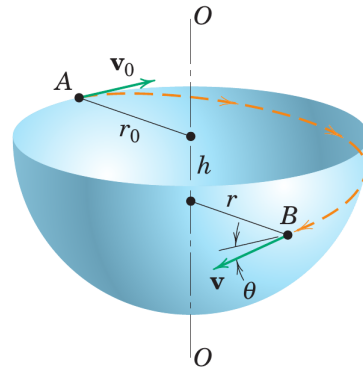


Figure 4

Question 4: A small mass particle is given an initial velocity v_0 tangent to the horizontal rim of a smooth hemispherical bowl at a radius r_0 from the vertical centerline, as shown at point A in Figure 4. As the particle slides past point B , a distance h below A and a distance r from the vertical centerline, its velocity v makes an angle θ with the horizontal tangent to the bowl through B . Determine θ .

Question 5: Three steel spheres of equal weight are suspended from the ceiling by cords of equal length which are spaced at a distance slightly greater than the diameter of the spheres (refer Figure 5). After being pulled back and released, sphere A hits sphere B , which then hits sphere C . Say, the coefficient of restitution between the spheres is denoted by e , and the velocity of A just before it hits B is v_0 , determine, (a) the velocities of A and B immediately after the first collision, (b) the velocities of B and C and immediately after the second collision. (c) Assuming that the n spheres are suspended from the ceiling and that the first sphere is pulled back and released as described above, determine the velocity of the last sphere after it is hit for the first time. (d) Use the result of part c to obtain the velocity of the last sphere when $n = 5$ and $e = 0.9$.

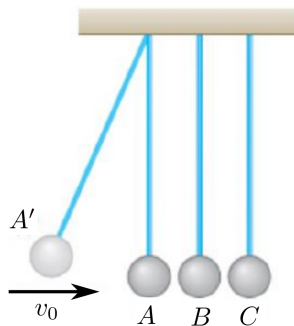


Figure 5

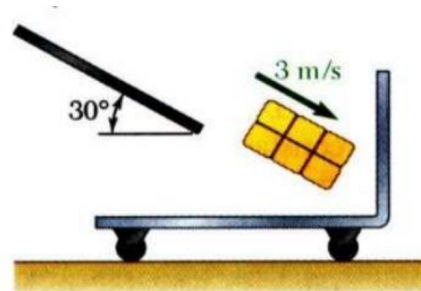


Figure 6

Question 6: As shown in Figure 6, a 10 kg package drops from a chute into a 20 kg cart with a velocity of 3 m/s. Knowing that the cart is initially at rest and can roll freely, determine, (a) the final velocity of the cart, (b) the impulse exerted by the cart on the package, (c) the fraction of the initial energy lost in the impact. Assume that there is no friction between cart and floor, and the distance from chute to wall of the cart is short, so that the package hits the cart at 3m/s and it is not like a projectile motion.